Measuring Intrahousehold Power in Multiple Domains

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Abstract

Measuring intrahousehold power is challenging because power may operate differently across domains of household decision-making. Using rich data from married couples in Bangladesh and three machine-learning methods, we study whether commonly used survey indicators of intrahousehold decision-making, agency, and control provide a domain-general proxy for women's wellbeing relative to their husbands. They do not. The indicators that best predict relative time use, consumption, and health differ substantially, and overlap among top predictors across domains is limited. These patterns are consistent with a multi-issue collective household framework in which bargaining weights and constraints may vary across domains, so that influence over one sphere of household decision-making need not translate into influence over others.

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1 Introduction

Measuring decision-making power and welfare within households is challenging. Household surveys often collect data at the household rather than individual level, making it difficult to observe intrahousehold dynamics or the decision-making process between spouses. This task is further complicated by the fact that intrahousehold power spans several spheres, from freedom of mobility and agency in time allocation to control over family resources.

Social scientists have proposed multiple ways to measure women's relative power and agency within families. These include widely used survey-based indices, such as the Women's Empowerment in Agriculture Index (WEAI; Alkire *et al.* (2013)) and the Survey-based Women's Empowerment Index (SWPER; Ewerling *et al.* (2017)). More recent approaches draw on machine learning and qualitative tools, such as the Machine Learning and Semi-structured Interviews (MASI) method of Jayachandran *et al.* (2023). Lab-in-the-field experiments have also been employed to improve measurement by eliciting willingness to pay for control over resources as a proxy for women's decision-making power (Almås *et al.*, 2018). Beyond empirical methods, theory-based approaches such as the collective household model (Chiappori, 1988, 1992; Browning *et al.*, 2013) provide a systematic framework for conceptualizing and analyzing women's power within the family. While each of these approaches has advanced the field, many empirical applications either emphasize a particular domain of power or aggregate diverse dimensions into a single index, potentially obscuring variation across spheres of household decision-making.[1]

Our study examines the multidimensional nature of intrahousehold power by evaluating how well questions from commonly used survey modules predict different aspects of women's wellbeing relative to their husbands.[2] Specifically, we investigate whether indicators associated with power in one domain are also informative about relative spousal wellbeing in other domains, such as consumption, health, and time use. Our primary objective is not to propose a new survey instrument, nor to isolate a structural bargaining parameter from each survey response. Instead, we assess whether existing questions commonly used to measure decision-making, agency, and control behave empirically as a unified measure of intrahousehold power or instead contain domain-specific information with distinct implications for individual wellbeing.

We draw on a comprehensive household survey from Bangladesh that combines rich questionnaires on women's and men's intrahousehold power with detailed measures of individual consumption, health, and time use. The survey includes questions on a wide range of topics, from participation in household economic decisions to experiences of domestic abuse. Importantly, many of these questions were asked of both husbands and wives, allowing us to assess not only which questions are most informative but also how responses and their predictive performance differ by respondent.

We begin by using machine learning methods to study whether the empirical content of power-related survey questions is common across domains or instead domain-specific. We posit two possibilities. The first is that the relevant survey indicators capture a largely unified dimension of in-

[1] One exception is the Women's Empowerment in Agriculture Index (WEAI), which is explicitly multidimensional, encompassing five domains: production, resources, income, leadership, and time (Alkire *et al.*, 2013; Quisumbing, 2025).

[2] While women's power extends beyond the household—political participation, representation, and broader societal influence—our analysis focuses on intrahousehold power and wellbeing.

trahousehold power. In that case, strength in one domain, such as decision-making over time use, would tend to reflect strength in others, such as control over household resources or agency over one's own health. Even if spouses' preferences differ by domain, so that relative outcomes may vary or even move in opposite directions, the same underlying power-related factors should systematically shape outcomes in all spheres. In this case, the same survey questions should consistently predict relative wellbeing across diverse dimensions. The second possibility is that the relevant dimensions of power, agency, information, and control are domain-specific. Power in one domain need not translate into power in another, and different aspects of wellbeing may be best predicted by different sets of indicators rather than by a single overarching measure.

Importantly, simple correlations in relative outcomes—or even in responses to power-related questions—cannot distinguish between these views. Our approach instead focuses on the mapping from survey indicators to spousal relative outcomes. By examining whether the same indicators consistently carry predictive signal across domains, we provide evidence on whether these commonly used measures behave as a common latent construct or as a set of domain-specific signals.

Our results align with a multidimensional view. Questions that strongly predict one outcome, such as relative time use, are often weak predictors of others, such as relative consumption or health. This pattern suggests that the survey measures commonly used to capture intrahousehold power contain substantial domain-specific information. It challenges the notion that a single index can summarize the aspects of intrahousehold power most relevant for all dimensions of individual wellbeing and underscores the need to tailor measurement tools to the particular dimension under study.

Our analyses employ a range of machine learning techniques. Our preferred approach uses the random forest algorithm, which ranks survey questions based on how informative their responses are for predicting multiple correlated target variables. For instance, when analyzing time use, we consider several related outcomes: women's market work, household work, and stated satisfaction with time use relative to their husbands. Similarly, for consumption, we include both women's caloric and micronutrient intakes and the monetary value of food consumed relative to their spouses. For health status, we consider measures of both short-term and long-term illnesses. To ensure robustness, we also apply alternative machine learning methods—such as multi-task Lasso and group Lasso stability selection—to assess the sensitivity of our findings to the choice of prediction algorithm.

Our results show that the survey questions whose responses are most predictive of intrahousehold relative outcomes vary substantially across domains. Based on the random forest algorithm, we find little to no correlation in feature importance rankings across time use, consumption, and health. For example, the correlation between feature importance in predicting relative time allocation and health is just 0.0935; the correlation between the predictive performance of responses to power-related survey questions for consumption and health is similarly low. Results from group Lasso and multi-task Lasso stability selection further reinforce this finding: among the top 10 survey questions for each outcome, there is no overlap across the three dimensions. Even when expanding to the top 50 questions, overlap remains minimal, with only 14 percent of questions identified as top predictors in more than one domain.

We conclude the paper by interpreting our empirical findings through a simple conceptual framework based on the collective household model (Chiappori, 1988, 1992; Browning *et al.*, 2013; Dunbar *et al.*, 2013). The framework allows bargaining weights and constraints to vary across decision domains, and recognizes that choices in one domain may affect choices in others. To generate meaningful domain-specific bargaining, the framework requires limits to costless renegotiation or to bargaining jointly over all domains—for example because of coordination costs, limited attention, institutional constraints, or decisions that are difficult to revisit once made. Under these conditions, household decision-making may be conditionally efficient within a given domain, taking other choices as fixed, without necessarily being globally efficient across all domains (Lewbel & Pendakur, 2022). The framework is not intended to identify domain-specific bargaining weights in the data. Rather, it provides one way to rationalize why survey indicators associated with intrahousehold power and agency may have different predictive content for relative wellbeing across domains.

Our main contribution is to incorporate machine learning techniques to study the measurement of intrahousehold power and its relationship to relative spousal wellbeing.[3] The focus on the multidimensionality of women's power builds on Kabeer (1999), who emphasizes the need for comprehensive measures capturing women's access to resources, power, and achievements, consistent with Sen's capabilities framework (Sen, 1990; Sen *et al.*, 1999). Laszlo *et al.* (2020) similarly argue that traditional approaches—centered on income and asset ownership—miss the complex dynamics of decision-making within households. Donald *et al.* (2020) make related points and propose a multidisciplinary framework to address these gaps. Relatedly, Lundberg & Pollak (1993) formalize a model where responsibilities and power are divided into separate spheres, shaped by social norms.

We also contribute to recent work on developing survey tools to measure women's power. Jayachandran *et al.* (2023) combine machine learning with qualitative measures to design a concise survey module. Our goals differ: rather than designing new instruments, we assess whether existing questions predict different observable aspects of women's wellbeing—specifically time use, consumption, and health—rather than qualitative measures from semi-structured interviews. In doing so, we ask whether questions that are informative in one domain are also informative in others.

The policy relevance of *unbundling* women's power is highlighted by Anderson (2022), who stresses that its dimensions interact and co-evolve in ways not yet fully understood. Policies often assume that greater power leads to broader wellbeing gains. Evaluating how well survey questions predict outcomes across domains is therefore essential. Our paper takes a step in this direction through both measurement and conceptual contributions.

2 A Simple Benchmark and Empirical Implications

Before describing our empirical exercise, we clarify the benchmark against which we interpret our results. The goal is not to estimate a structural household model, but to formalize the empirical implications of a domain-general measure of intrahousehold power.

Consider a household with two spouses, indexed by $j \in w, m$, and two domains of decision-making, indexed by $d \in 1, 2$. These domains may correspond, for example, to consumption and time use, or to time use and health investments. We present the two-domain case for notational simplicity;

[3] Jayachandra & Voena (2025) provide an insightful review of both the measurement and theory of women's intrahousehold power in low-income settings.

the same logic extends directly to any finite number of domains.

In the standard collective household model, household allocations are Pareto efficient and can therefore be represented as the solution to a weighted sum of individual utilities, with Pareto weights summarizing the bargaining power of household members (Chiappori, 1988, 1992; Browning *et al.*, 2013; Dunbar *et al.*, 2013). Let a_{jd} denote spouse j's allocation in domain d. In a two-domain version with a common bargaining environment, the household program is

$$\max_{a_{wd},a_{md}} \mu U_w(a_{w1},a_{w2}) + (1-\mu)U_m(a_{m1},a_{m2}) \quad \text{subject to} \quad (a_{w1},a_{m1},a_{w2},a_{m2}) \in A(z), \tag{1}$$

where $\mu \in [0,1]$ is the Pareto weight on the wife, $1-\mu$ is the Pareto weight on the husband, and $A(z)$ is the household feasible set.

Pareto efficiency implies that, given the household feasible set, there are no unrealized reallocations across spouses or domains that could make one spouse better off without making the other worse off. In the benchmark above, the same Pareto weight μ governs allocations across both domains. Realized relative outcomes may still differ because preferences, prices, technologies, or constraints differ by domain, but the underlying bargaining environment is summarized by the same scalar weight.

This benchmark has a direct implication for how we think about survey measurement. Let $X = (X_1,\ldots,X_p)$ denote the set of survey questions used to measure intrahousehold decision-making power, and let Y_d denote a relative spousal outcome in domain $d \in 1,2$. If a common bargaining environment governs both domains, then the same underlying scalar measure of relative influence should be relevant for outcomes in each domain. The Pareto weight μ is not directly observed, but if the survey questions contain information about this common bargaining environment, that information should be summarized by a common scalar object. Let $S(X)$ denote this summary—for example, the component of the common bargaining environment that can be inferred from the survey responses. A single-index representation then takes the form

$$E[Y_d \mid X] = g_d(S(X)), \qquad d \in 1,2, \tag{2}$$

for some domain-specific functions $g_1(\cdot)$ and $g_2(\cdot)$. These functions may differ across domains, allowing the same bargaining environment to translate into different relative outcomes. Thus, relative outcomes in the two domains need not move together, and the strength of their association with the common bargaining environment may differ. The key restriction is that the information contained in X about relative influence is summarized by the same scalar $S(X)$ in both domains.

Equation (2) motivates a set of empirical predictions. If a common bargaining environment governs both domains, then survey indicators that contain information about relative power should tend to carry predictive information in both domains. We would therefore expect some positive association in feature-importance rankings and meaningful overlap among the highest-ranked predictors across the two domains. We would also expect appropriately constructed aggregate measures of the common signal to retain predictive content for both outcomes.

The alternative is that the bargaining environment itself differs across domains. In that case, the information contained in the survey responses about relative power need not be summarized by a single scalar $S(X)$, but rather by a pair of domain-specific components,

$$S(X) = (S_1(X), S_2(X)),$$

with

$$E[Y_1 \mid X] = g_1(S_1(X)), \qquad E[Y_2 \mid X] = g_2(S_2(X)). \tag{3}$$

Under this view, a survey question may contain substantial information about the bargaining environment relevant for one domain and little information about that relevant for the other. We would therefore expect weaker correspondence in predictor rankings, limited overlap among top predictors, and poor performance of simple aggregate measures that impose a common weighting of survey indicators across the two domains.

Our empirical exercise evaluates these implications. We do not estimate the Pareto weight μ, recover the latent functions $S_1(\cdot)$ and $S_2(\cdot)$, or test Pareto efficiency. Instead, we examine whether commonly used power-related survey questions contain a common signal about intrahousehold power across different domains or whether their predictive content is domain-specific. We do so by comparing feature-importance rankings, measuring overlap among top predictors, and assessing how much predictive power is retained when the full predictor set is compressed into simple common indices.

3 Data

We use data from the 2018–2019 Bangladesh Integrated Household Survey (BIHS) conducted by IFPRI. The survey includes rich measures of intrahousehold power alongside detailed information on health, individual consumption, and time use. We focus on 1,620 rural households with married couples (with or without children), excluding those with multiple adult men or women to avoid more complex family structures.

3.1 Outcomes

Our objective is to predict women's wellbeing relative to their spouses, encompassing multiple dimensions that are often challenging to measure. The design of BIHS allows us to address these challenges, as it collects detailed information on time use, health, and individual consumption for both women and men. Table A2 in the Appendix lists the outcome measures used in our analysis.

Time use is measured in hours spent on income-generating and domestic work. Respondents completed a 24-hour diary, recording activities in 15-minute intervals across 23 categories such as "sleeping," "personal care," and "farming." For our analysis, these are aggregated into leisure, domestic, and income-generating work. Alongside these objective measures, we construct an index of subjective satisfaction based on survey questions asking respondents to rate, from 1 ("not satisfied") to 10 ("very satisfied"), aspects such as the division of household work and leisure time. We combine these ratings using principal component analysis to create a composite satisfaction index.

In our sample, households allocate over two-thirds of their budget to food, on average. We therefore use relative food consumption as a meaningful proxy for relative intrahousehold consumption.

Food quantity is assessed through caloric intake and the monetary value of total food consumption, while food quality is evaluated on the basis of nutrient intake, including protein, iron, vitamins, and zinc. These measures are constructed from a detailed survey module that recorded individual food consumption over the course of a single day. To generate these data, enumerators interviewed the household member primarily responsible for food preparation, collecting information not only on the meals cooked and ingredients used, but also on which household members consumed each meal.[4]

To construct health outcomes, we use principal component analysis to summarize survey questions on long-term disabilities and short-term morbidities. The first set captures functional limitations—such as difficulties standing, walking long distances, carrying loads, or with hearing, speech, and eyesight—reflecting chronic conditions that affect participation in daily activities. The second set focuses on recent illnesses (past four weeks), including fever, diarrhea, and cough, as well as diagnosed conditions like malaria or pneumonia, alongside workdays lost.[5] We also compute body mass indices (BMI) as a complementary health indicator.[6] Finally, we construct an index of domestic abuse based on reports of threats of divorce, and verbal or physical violence.

Since our analysis focuses on intrahousehold relative wellbeing between spouses, we transform the outcome variables described above—with the exception of the domestic violence index, which is not measured for men—by dividing women's values by the sum of women's and men's values. The distributions of these relative outcomes across different domains are shown in Figure A2 in the Appendix.

3.2 Predictors

In Table A1 in the Appendix, we summarize the survey modules used to measure relative intrahousehold power and the number of questions drawn from each. These questions serve as possible predictors of relative spousal outcomes discussed above. The "women's status" component captures the degree to which women are allowed to work or face opposition from husbands or in-laws. It also records whether women retain, share, or surrender control over their earnings (if any), how decisions on everyday expenses or loans are made, and whether women themselves can make such financial decisions. It also tracks freedom of movement to markets, clinics, or social events, as well as autonomy in reproductive choices. The WEAI module complements this information by shifting the focus to economic and civic agency more broadly. It probes women's decision roles in farming, livestock, business, and wage employment, and whether they can influence how income and assets are used. Ownership and control over land, equipment, housing, and other durable goods are documented alongside access to loans and household spending.[7]

Note that the sets of predictors by design differ by gender. While the majority of questions are

[4]Several precautions were taken to enhance reliability and reduce measurement error. Details are provided in D'Souza & Tandon (2019) and Brown *et al.* (2021).

[5]Since PCA indices can take negative values, we rescale the values using an inverse logit function so they are between zero and one. Additional details on the data construction are provided in Section C in the Appendix.

[6]BMI can also be viewed as a nutrition outcome. As a robustness check, we classify it under food quantity and find similar results.

[7]In addition, the WEAI module includes questions that may seem only peripherally related to intrahousehold power. In particular, it asks respondents to reflect on the motivations behind their actions in multiple domains—such as work, household expenditures, health, religious expression, and family planning—and to rate the extent to which their behavior is driven by external incentives (punishment or rewards), social expectations (reputation, avoiding blame), or personal values and interests. It also records satisfaction with decision-making in these areas.

asked to both men and women (that is, to both spouses within the same couple), certain modules, such as the one focused on freedom of mobility, are administered only to women. Further differences in the number of available predictors arise because questions with no variation in responses are excluded from the analysis. In specifications that pool men's and women's predictors, a predictor is an individual response, which is essentially defined as the interaction between a survey question and the sex of the respondent.[8]

All questions in these modules are closed-ended and include both binary and multiple-choice formats. Some questions require a simple "yes" or "no" response, such as "Do you alone have any money that you can decide how to spend?" Others present multiple options, such as "Who made the decision to use birth control?" with possible answers including "yourself," "your husband," "both yourself and your husband," or "someone else." For questions with categorical responses that lack an inherent ranking, we re-code the answers into multiple indicator variables. Consequently, the survey questions translate into a total of 1,078 predictor variables across men and women.[9,10]

4 Methodology

This section outlines our approach to selecting survey questions that best predict spousal relative outcomes. We frame the exercise as a multi-task prediction problem, allowing the model to predict several related outcomes within each domain jointly rather than estimating a separate model for each outcome.[11] Our main specifications pool survey responses from women and men, but we also estimate models separately by respondent gender.

4.1 Random Forest

Our preferred approach employs a random forest regressor, an ensemble learning method that constructs multiple decision trees on bootstrapped samples of the training data. Decision trees partition the data by recursively applying decision rules that minimize mean squared error (i.e., within-node variance). Because individual trees tend to overfit, random forests introduce two sources of randomness—bootstrapping the data and selecting a random subset of features at each split—which improves generalization.

To assess the predictive performance of power-related survey questions, we use permutation feature importance (Breiman, 2001), which measures the decrease in predictive accuracy when a given feature is randomly shuffled—features whose shuffling leads to a larger drop in accuracy are those with greater predictive power. This approach mitigates biases in impurity-based importance and

[8] Prior research by Ghuman *et al.* (2006) and Ambler *et al.* (2021) document significant discrepancies between spouses in their responses to questions about empowerment. The findings of Ambler *et al.* (2021) are particularly relevant, as their study also utilizes the BIHS and reveals substantial disagreement regarding decision-making authority and asset ownership within households.

[9] Specifically, the 172 survey questions yield 591 predictor variables for women, while the 143 questions yield 487 predictor variables for men. In certain specifications, we restrict our algorithm to select questions (i.e., groups of predictors) rather than individual predictors. This is discussed further in Section 4.

[10] Not all survey questions are asked of every respondent due to skip sequencing or conditional branching (see Manski & Molinari, 2008). For example, only landowners are asked who decides how land income is spent. To retain such questions without dropping non-landowners, we merge initial and follow-up questions into composite variables—for instance: "Do you own land, and if so, who decides how income is spent?" with response options capturing own, spouse, joint, or not applicable. All predictors are thus recoded for consistency and coverage.

[11] Further technical details are provided in Appendix A.

yields robust rankings. We repeat the procedure 100 times on random 50 percent subsamples of the data and use five-fold cross-validation to tune hyperparameters. The most relevant features are ranked by their average decrease in predictive accuracy across iterations.

4.2 Alternative Approaches

We test robustness using alternative machine learning algorithms. Specifically, we employ multi-task Lasso (Argyriou *et al.*, 2006), which extends standard Lasso to select predictors for multiple related outcomes simultaneously. Its shared sparsity constraint ensures that the same features are selected across related outcomes. We combine multi-task Lasso with stability selection (Meinshausen & Bühlmann, 2010): in each of 1,000 iterations, we apply it to a random 50 percent subsample, use five-fold cross-validation to choose the regularization parameter, and rank predictors by selection frequency.[12]

As discussed in Section 3, survey questions with multiple categorical responses are converted into multiple binary indicators. To address this specific feature of the data, we also use group Lasso (Yuan & Lin, 2006) to ensure that entire questions, rather than individual binary indicators, are selected jointly. This method imposes a structured sparsity constraint at the group level by penalizing all coefficients belonging to the same question together, thereby enforcing group-wise selection. As with multi-task Lasso, we integrate group Lasso with stability selection, applying the method iteratively to subsamples of the data. Predictors are then ranked based on their selection frequency.

5 Results

This section presents three sets of results. We first examine the main empirical implication of the common-bargaining-environment benchmark presented in Section 2, by asking whether the same survey questions carry predictive information about relative spousal outcomes across domains. We then classify the most predictive survey questions into broader domains of decision-making power and agency to assess whether conceptually related forms of relative influence predict outcomes within or across domains. Finally, we study predictive performance directly: how much variation the selected survey questions explain, how predictive content differs by respondent gender, and how much information is retained when the full set of questions is compressed into a single common index.

5.1 Domain-Specific Predictive Content

We begin by ranking survey questions about intrahousehold power according to their feature importance in predicting relative spousal outcomes across three domains: time use, consumption, and health. These rankings are generated using the random forest algorithm described in the previous section. To assess whether the same questions are informative across domains, we compare their importance rankings across outcome groups. Because feature-importance scores are not directly comparable in magnitude across models, we focus on rank correlations.

[12] We conduct sensitivity checks using standard Lasso stability selection, which does not require the same features be selected across outcomes within a dimension. Our results are confirmed and available upon request.

These rank correlations summarize the similarity in which survey questions are predictive across domains; they are not correlations among the underlying relative outcomes. For example, even if women's relative health and relative time use were positively correlated, the survey questions that best predict one outcome need not be the same questions that best predict the other. Conversely, even weakly correlated outcomes could be predicted by similar survey questions if they reflect common dimensions of intrahousehold decision-making, agency, or control.

Figure 1 illustrates the correlation—or more accurately, the lack of correlation—in rankings of predictors across domains. If the same set of survey questions provided a stable domain-general summary of women's relative wellbeing, we would expect feature-importance rankings to be similar across consumption, health, and time use. Instead, the scatter plots show little alignment with the 45-degree line. This pattern suggests that the survey questions that are informative for one relative outcome domain carry limited information about the others.

Pairwise rank correlations are small and sometimes negative. The rank correlation between the importance of survey questions in predicting women's relative time allocation and health status is 0.0935. The correlation between the importance scores for consumption and health is -0.0444 and is not statistically significant. The correlation between time use and consumption is -0.1399 and is statistically significant. These patterns are consistent with the view that survey questions commonly used to proxy for intrahousehold power have domain-specific predictive content.

The same conclusion emerges when we compare the highest-ranked survey questions across domains. Figure 2 shows overlap among the top 10, top 50, and top 100 predictors. None of the top 10 predictors of relative outcomes in time allocation, consumption, and health is shared across all three domains. No single survey question appears among the top 50 predictors in all three domains. Overlap remains limited even pairwise, and expanding the set to the top 100 predictors yields only three questions common to all domains.[13]

A broadly consistent picture emerges from group Lasso and multi-task Lasso stability selection. Among the top 10 predictors of women's relative outcomes in time allocation, health, and consumption—defined by frequency of selection across 1,000 iterations of stability selection—there is no overlap across domains under either method. As the predictor set expands, some overlap emerges mechanically, but it remains modest. Among the top 50 predictors, only about 14 percent are identified as strong predictors in more than one domain by both multi-task Lasso and group Lasso.

Finally, we replicate the analysis restricting the set of predictors to survey questions asked exclusively to women. The results closely mirror those above. Predictor rankings are again essentially uncorrelated across domains, with rank correlations close to zero: for instance, 0.0171 for time use versus health and 0.0498 for consumption versus health. Overlap across domains is also minimal: no common predictors among the top 10, one among the top 50, and six among the top 100.

5.2 Cross-Domain Predictive Patterns

The preceding section shows that the same individual survey questions rarely appear among the top predictors across all outcome domains. We next ask a related but distinct question: whether the most

[13] For completeness, we provide the list of ten most predictive survey questions across each of our three outcome domains for the pooled women's and men's survey responses. The results using random forest are presented in Tables A3, A4, and A5 in the Appendix.

predictive questions nevertheless come from similar conceptual areas of household decision-making. This distinction is useful because two indicators may differ at the question level while still reflecting a common underlying sphere of agency or control. For example, a question about who decides how food is purchased and a question about who controls income used for household expenses are distinct survey items, but both may reflect decision-making over consumption resources.

To examine this broader structure, we classify the top 50 predictors from the random forest analysis into thematic groups, which we refer to as *power domains*. These groups summarize the substantive content of the survey questions rather than their statistical ranking. They include domains such as consumption, time use, health, assets, mobility, fertility, savings and credit, agricultural production, and domestic violence. This aggregation allows us to assess whether predictors that are conceptually tied to one sphere of decision-making are informative only for outcomes in the same sphere, or whether they also predict relative wellbeing in other domains. To validate this classification, Appendix B compares our manual groupings with those obtained through text-embedding clustering and finds strong alignment.

Figure 3 maps the 50 most predictive survey questions to relative outcome domains. The left side of the figure reports the power domain from which each top predictor originates, while the right side reports the domain of relative spousal outcomes it predicts. The width of each flow indicates the share of top predictors in an outcome domain that come from a given power domain. Less common categories—such as fertility, agricultural production, savings and loans, mobility, and non-farm asset decisions—are grouped under "Other" to preserve readability.

Two patterns emerge. First, conceptually related questions about intrahousehold power often predict outcomes outside their own domain. For instance, some of the top predictors of health outcomes come from questions about consumption and time use. This does not imply that the same individual survey questions are predictive across domains; the previous section showed that overlap at the question level is limited. Rather, it suggests that broader spheres of household decision-making may be linked across domains. Control over time, resources, or household expenditures may shape not only the corresponding outcome but also other dimensions of relative wellbeing.

Second, own-domain predictors remain important. Consumption-related questions are especially predictive of consumption outcomes, health-related questions are important for health outcomes, and time-use questions account for a substantial share of the top predictors of time-use outcomes. This pattern is consistent with the idea that power-related survey questions contain domain-specific information: questions tend to be most informative for outcomes that are conceptually close to the decision sphere they measure, even though cross-domain links are also present.

Overall, these patterns reinforce the main result while adding nuance. The low overlap in individual top predictors does not mean that the survey indicators are idiosyncratic or unstructured. Instead, predictive content appears organized around broader domains of decision-making, with both own-domain relevance and cross-domain connections.

5.3 Predictive Performance and Respondent Choice

The preceding results show that the survey indicators with the greatest predictive content differ substantially across domains. We next ask how much predictive power these indicators contain, whether this predictive content differs by respondent gender, and how much of it can be summarized by a single aggregate index.

Figure 4 plots R^2 from linear regressions that incrementally add the top k predictors selected by the random forest, ordered by feature importance. The level of R^2 captures overall predictive power, while the shape of each curve shows how quickly predictive content accumulates as additional survey indicators are included. To provide a benchmark for scale, Table A2 in the Appendix reports the R^2 from regressions of the same relative outcomes on standard demographic controls: spouses' education, age, age squared, literacy, and the number of children in the household. These demographic controls explain relatively little variation in most outcomes, with average R^2 values of about 0.05 for time use, 0.02 for consumption, and 0.02 for health. Against this benchmark, the selected power-related indicators contain meaningful predictive information, especially for time-use outcomes.

Time-use outcomes are the most predictable: market and domestic work measures reach R^2 values around 0.15–0.20, with most of the gains coming from the top 20 predictors. Consumption outcomes display more moderate and gradual gains, with explanatory power generally remaining below 0.10. Health outcomes are more heterogeneous: domestic violence is relatively predictable, while other health measures are harder to explain using these survey indicators. Across domains, the curves are generally concave, indicating that much of the predictive content is concentrated in a relatively small subset of questions.

We also examine whether predictive content differs depending on whether survey questions are answered by women or men. Appendix Figure A4 shows that women's responses account for a larger share of the most predictive indicators across domains. At the same time, men's responses are not redundant: models that combine men's and women's responses have higher R^2 than models based on either respondent alone. These results have direct implications for survey design. When only one spouse can be interviewed, women's responses appear to provide stronger predictive content for relative wellbeing outcomes. When feasible, however, interviewing both spouses is valuable because men's responses provide complementary information about intrahousehold power.

Finally, we examine another empirical implication of the common-bargaining-environment benchmark developed in Section 2. That benchmark implies that, if the information contained in the survey responses about intrahousehold power can be summarized by a common scalar $S(X)$, then a suitably constructed common index should retain predictive content across relative outcomes. As simple empirical counterparts to $S(X)$, we construct two indices using all survey questions included in the predictor set: an equal-weighted average of the power indicators and their first principal component.[14] Figure A5 in the Appendix reports the R^2 from regressions of each relative outcome on these indices.

[14] Before constructing the indices, we standardize each variable to have mean zero and unit variance. This prevents survey questions measured on larger scales or with greater variance from receiving mechanically greater weight in either the equal-weighted average or the principal-component index.

The predictive power of both indices is quite low. Across domains, they explain less than two percent of the variation in relative spousal wellbeing, roughly comparable to or below the explanatory power of standard demographics for most domains. This contrasts with the top-k selected-predictor results in Figure 4, where allowing survey questions to enter separately yields substantially higher explanatory power, especially for time-use outcomes. The comparison suggests that the predictive information contained in the survey responses is not well summarized by these simple common indices.

6 A Multi-Issue Collective Household Model

Section 2 considered two domains for expositional simplicity and formalized the empirical implication of a common bargaining environment. In that benchmark, a single Pareto weight governs allocations across domains. One implication is that survey questions that contain information about intrahousehold power should therefore tend to carry a common predictive signal across relative spousal outcomes. The empirical results provide little support for this implication: predictor rankings differ substantially across domains, overlap among top predictors is limited, and simple common indices retain little predictive information.

We now generalize that setup to an arbitrary finite number of decision domains. The key departure from the benchmark is that households need not bargain jointly over all domains or costlessly renegotiate previously settled decisions. We therefore allow domains to be resolved sequentially, with the Pareto weights and constraints relevant for each decision potentially differing across domains. The framework is intended as an economic interpretation of the empirical results rather than as a structural model that we estimate.

Let there be two spouses, indexed by $j \in \{w, m\}$, and $D > 1$ decision domains, indexed by $d \in \{1, \dots, D\}$. These domains may include consumption, time allocation, health investments, fertility, mobility, savings, or asset ownership. Issues are resolved according to an agenda that determines decision order (Fershtman, 1990). Without loss of generality, suppose the domains are resolved in the order $1, \dots, D$. This sequential structure need not be dynamic: all decisions may occur within a single period. What matters is that households cannot costlessly bargain over all domains jointly or continuously reopen previously settled decisions. Such limits to renegotiation may arise from coordination costs, limited attention, institutional constraints, delegated responsibilities, or other frictions that make joint bargaining over all domains impractical.

For each domain $d \in \{1, \dots, D\}$, let

$$a_d = \{a_{jd}\}_{j \in \{w,m\}}$$

denote the allocation associated with that domain, where a_{jd} is the allocation received or chosen by spouse j in domain d. Let

$$a_{<d} = (a_1, \dots, a_{d-1})$$

denote the history of allocations resolved before domain d.

At each stage d, the household chooses the domain-d allocation a_d, taking previously resolved allocations $a_{<d}$ as given. The domain-d problem is

$$\max_{\{a_{jd}\}_{j\in\{w,m\}}} \sum_{j\in\{w,m\}} \mu_{jd}(z_d) U_{jd}(a_{jd}, a_{<d}) \quad \text{subject to} \quad a_d \in A_d(a_{<d}, z_d), \qquad d = 1, \ldots, D. \tag{4}$$

Here, U_{jd} denotes spouse j's utility from the allocation in domain d. For $d \geq 2$, utility may depend on previously resolved allocations $a_{<d}$, allowing choices in earlier domains to affect the value of allocations in the current domain. For example, time-allocation decisions may affect the returns to health investments, while control over income may affect the value of subsequent consumption choices.

The term $\mu_{jd}(z_d)$ is the Pareto weight attached to spouse j in domain d. A higher Pareto weight implies greater influence over the allocation chosen in that domain and, all else equal, a more favorable outcome for that spouse.[15] The vector z_d collects exogenous variables that shape the domain-d bargaining problem, including prices, household resources and endowments, institutional constraints, and distribution factors (i.e., variables that affect intrahousehold bargaining without directly affecting preferences or the household budget constraint; Browning *et al.* (2011)).

The key distinction from the benchmark in Section 2 is that the Pareto weights are indexed by domain. A spouse who has greater influence over one decision need not have the same influence over another. An individual's bargaining position is therefore summarized by a vector of domain-specific weights,

$$\boldsymbol{\mu}_j = (\mu_{j1}, \ldots, \mu_{jD}),$$

rather than by a single scalar measure of relative influence.

This multi-issue structure nests the single-weight benchmark in Section 2 as a special case. If all D domains are chosen jointly under a common set of Pareto weights, the household problem can be represented as a single collective optimization problem over the full vector of allocations. Cross-domain interactions in preferences, technologies, or constraints may still be present, but they are internalized when the household chooses all domains jointly. In this case, a scalar measure of relative influence may summarize the bargaining environment across domains. The framework here relaxes this restriction by allowing the Pareto weights and constraints relevant for each decision to be domain-specific and by allowing previously resolved choices to shape subsequent utility and feasible sets.

6.1 Conditional Efficiency

Sequential decision-making does not require noncooperative behavior within each domain. At each stage, the household may choose an allocation that is Pareto efficient conditional on previously resolved choices, the current feasible set, and the domain-specific Pareto weights. In domain d, the

[15] We do not allow the Pareto weights to depend on previously chosen allocations. Doing so would introduce an additional feedback mechanism in which household choices affect subsequent bargaining power, as in Basu (2006). In such a setting, earlier allocations may be chosen partly for their effect on future bargaining positions, introducing strategic considerations and potentially a different equilibrium concept. This mechanism is not needed for our purpose, which is simply to allow the bargaining environment to differ across decision domains. We therefore let μ_{jd} depend on exogenous features of the domain-specific bargaining environment, z_d, while allowing prior allocations $a_{<d}$ to affect subsequent preferences and feasible sets.

household chooses a_d efficiently conditional on $a_{<d}, A_d(a_{<d}, z_d)$, and

$$\{\mu_{jd}(z_d)\}_{j\in\{w,m\}}.$$

We refer to this as *conditional* or *issue-wise* efficiency.

Conditional efficiency need not imply global Pareto efficiency across all D domains. If earlier choices affect later utilities or feasible sets and cannot be costlessly reopened, the sequence of conditionally efficient decisions need not coincide with the allocation that would be chosen if all domains were negotiated jointly. Global efficiency is recovered if the household can bargain over all domains simultaneously while internalizing these cross-domain effects, or if earlier decisions do not affect the preferences or constraints relevant for subsequent domains. This notion of conditional efficiency closely parallels Lewbel & Pendakur (2022), with previously resolved issue-specific choices serving as the conditioning element.

6.2 Implications for Measurement

The framework provides one economic interpretation of the empirical patterns documented above. Under the common-bargaining-environment benchmark, survey indicators that contain information about relative influence should tend to carry a common predictive signal across domains. In the multi-issue framework, by contrast, the bargaining weights and constraints relevant for one domain may differ from those relevant for another. A survey question may therefore contain substantial information about relative influence in one domain while carrying little information about relative influence in another.

This distinction maps into our empirical exercise. Domain-specific bargaining environments can generate weak correspondence in feature-importance rankings and limited overlap among top predictors across domains, even when the survey indicators contain meaningful predictive information. They can also help explain why allowing individual indicators to enter separately yields substantially more predictive power than compressing the full predictor set into a common aggregate index.

It is important to note that this framework is not the only structure capable of generating these patterns. Alternative models of intrahousehold decision-making can also generate domain-specific bargaining power and are consistent with our empirical patterns. For example, Lundberg & Pollak (1993) propose a separate-spheres model in which social norms divide decision authority across domains. Issue-by-issue bargaining models, with or without asymmetric information, similarly allow influence to vary across issues (see, for example, Bac & Raff, 1996; Busch & Horstmann, 1999; Lang & Rosenthal, 2001). Our purpose is more limited: to show that the domain-specific predictive content documented in the data can arise within a collective household framework once the restrictions of a common bargaining environment across domains or of complete separability across decision domains are relaxed.[16]

[16] Much of the recent empirical collective-household literature isolates a particular domain of household choice. For example, Browning *et al.* (2013) and Dunbar *et al.* (2013) focus on the allocation of consumption within the household, effectively allowing the consumption problem to be studied separately from choices in other domains. By contrast, Chiappori & Mazzocco (2017) provide an example of a formulation in which the same Pareto weight governs joint decisions about consumption and time use. Our framework relaxes both of these polar cases: decisions need not be separable across domains, but neither must the same bargaining weights govern every domain.

There are several advantages to working within the collective household model. The efficiency assumption at the core of the collective model (Chiappori, 1988, 1992) offers substantial analytical advantages. First, it allows researchers to abstract from the particular bargaining protocol generating observed allocations. Second, efficiency implies decentralization properties that facilitate identification of key intrahousehold parameters, such as bargaining weights and the extent of joint consumption, when combined with additional structure (see, e.g., Browning *et al.*, 2013; Dunbar *et al.*, 2013; Bargain & Donni, 2012). In our framework, decisions may be efficient within domains conditional on choices made in other domains, while the resulting allocation need not be globally Pareto efficient across domains. Even in settings with this form of conditional rather than global efficiency, Lewbel & Pendakur (2022, 2024) show that important features of intrahousehold decision-making may remain identifiable. Structural identification and estimation of the multi-issue model are beyond the scope of this paper, but extending the analysis in this direction is a promising avenue for future work.

7 Conclusion

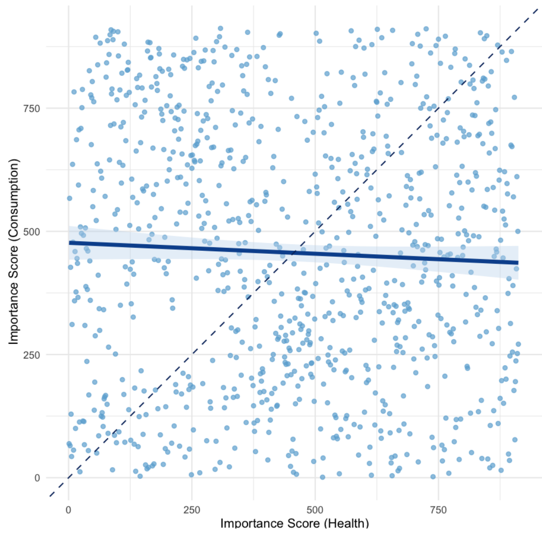
This paper studies how survey questions commonly used to capture intrahousehold power, agency, and control predict women's wellbeing relative to their husbands. Using rich survey data from married couples in Bangladesh and a suite of machine-learning methods, we show that these questions differ markedly in their predictive content across domains—time use, consumption, and health. Questions that are highly predictive in one domain often perform poorly in others, suggesting that measures of intrahousehold power may capture a bundle of domain-specific mechanisms rather than a single construct.

These patterns can be interpreted through the lens of a simple multi-issue collective household model. In the framework, household decisions are made across domains, and the bargaining weights and constraints relevant for one domain need not coincide with those relevant for another. Domain-specific weights may reflect differences in information, social norms, institutional constraints, or delegated responsibilities. As a result, influence over one sphere of decision-making need not translate into influence over others, and survey indicators linked to one domain may have limited predictive content for relative outcomes in another.

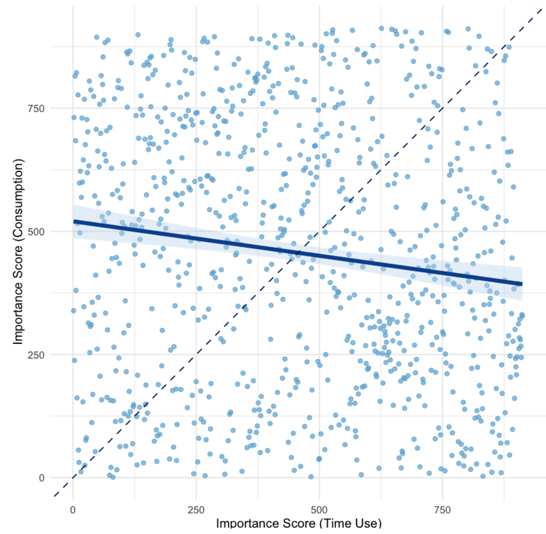
These findings have implications for both measurement and policy. Programs that seek to strengthen women's intrahousehold power often presume broad, cross-domain gains. Our results suggest that such gains should not be assumed. Researchers and policymakers should therefore measure the relevant domain of wellbeing directly and tailor interventions to the dimensions of agency and control most closely connected to the outcomes they aim to improve.

Figures

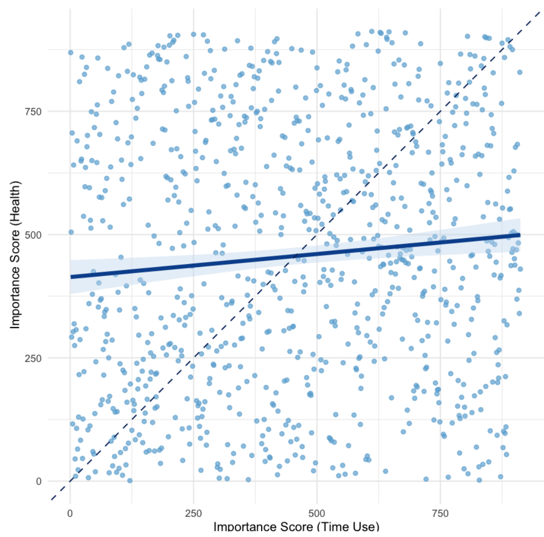
Figure 1: Correlation in Feature Importance of Survey Indicators



(a) Consumption vs. Health



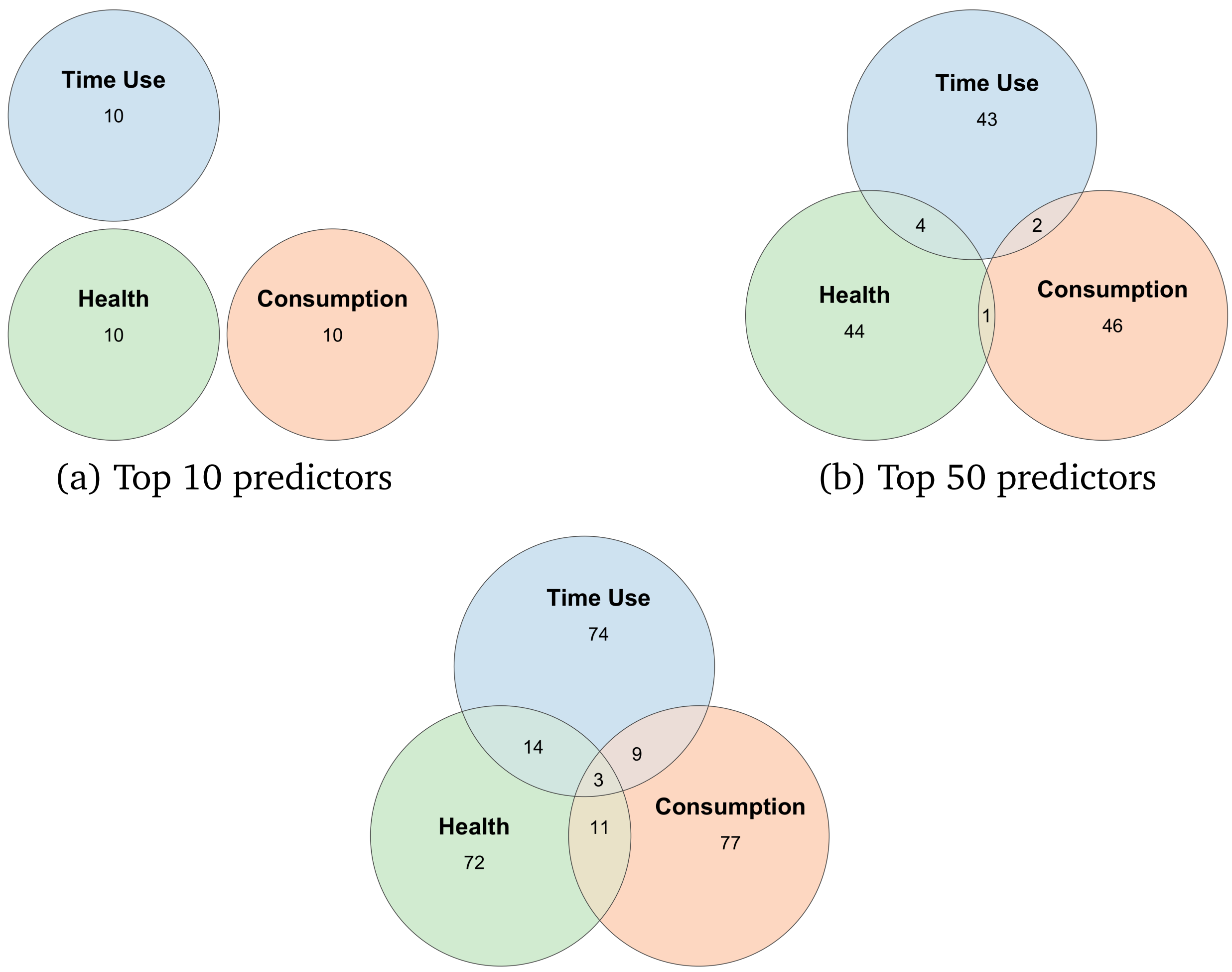
(b) Consumption vs. Time Use



(c) Health vs. Time Use

NOTES: Relationship in the ranking of predictors across three domains of relative spousal wellbeing using random forest permutation feature importance. Panel (a) compares consumption and health, panel (b) compares consumption and time use, and panel (c) compares health and time use. The 45-degree line indicates perfect agreement in rankings across domains.

Figure 2: Overlap of Top Predictors Across Domains

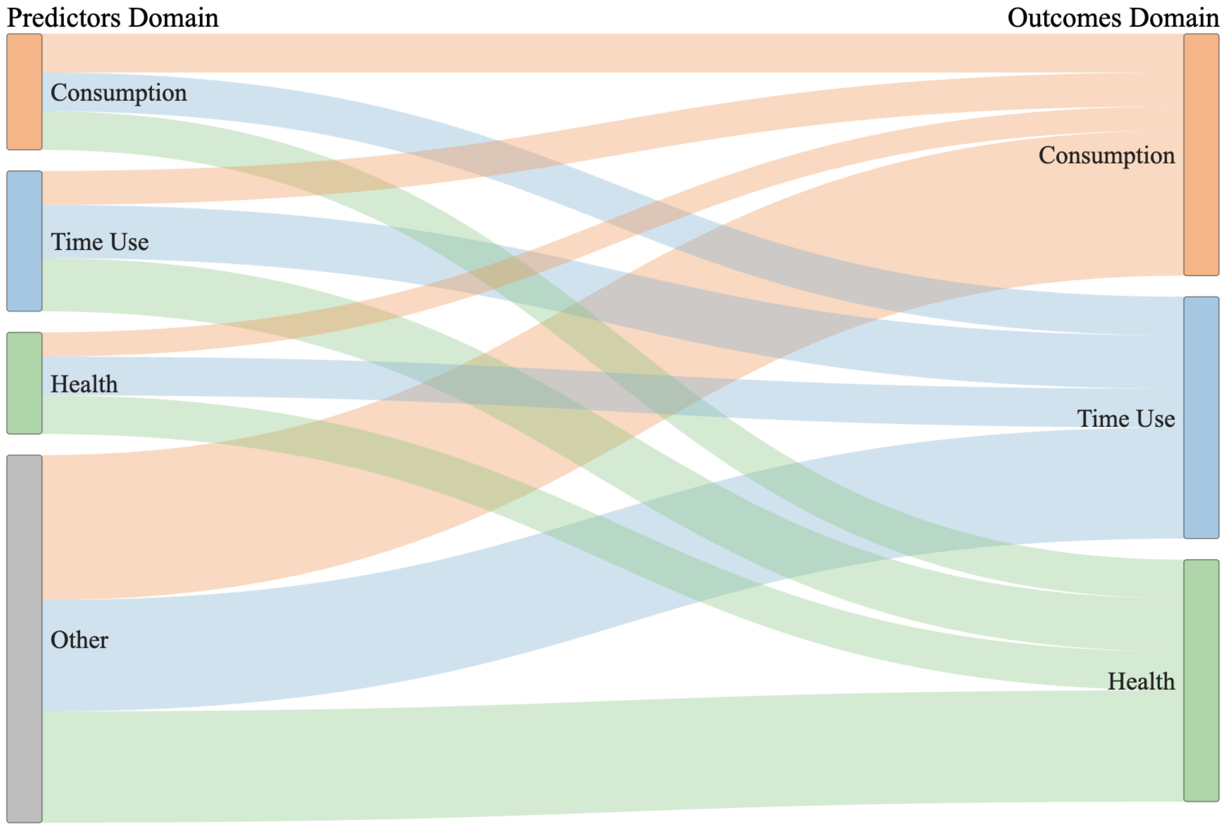


(a) Top 10 predictors (b) Top 50 predictors

(c) Top 100 predictors

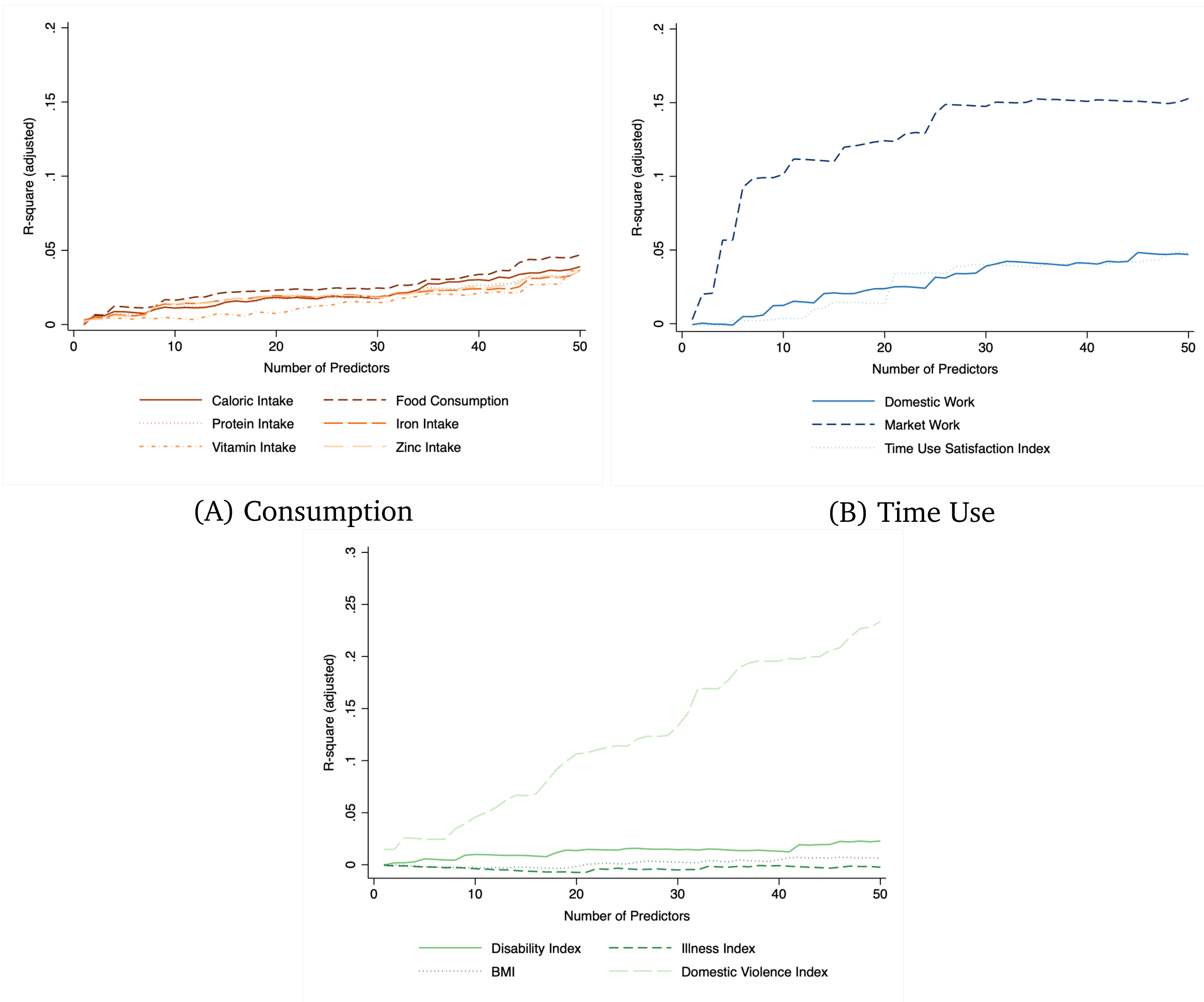
NOTES: Venn diagrams showing the overlap between top 10, top 50, and top 100 predictors across the three domains of relative spousal wellbeing based on random forest permutation feature importance scores.

Figure 3: Predictor Domains and Relative Outcome Domains



NOTES: The Sankey diagram illustrates the distribution of the top 50 survey questions ranked by random forest permutation feature importance across predictor domains and outcome dimensions. The width of each flow represents the share of variables from a given predictor domain that are among the top 50 predictors of relative spousal wellbeing in consumption, time use, and health. The "Other" category includes predictors related to fertility, religious expression, freedom of mobility, agricultural production excluding time use, savings and loans, and non-farm asset decisions.

Figure 4: Predictive Power of Survey Indicators



(A) Consumption

(B) Time Use

(C) Health

NOTES: This figure shows predictive performance for relative spousal wellbeing outcomes. For each domain, we plot the adjusted R^2 from linear regressions that incrementally add the top k survey indicators, ranked by random forest feature importance. Predictors are survey-question responses interacted with respondent gender.

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Supplemental Appendix

Contents

A Methodology: Details

In this section, we describe our approach to selecting survey questions that are the most predictive of women's relative outcomes within a household. We perform this analysis separately for questions asked only to women, only to men, and for the combined men's and women's questionnaires. Additionally, we perform the analysis separately for each welfare dimension. Since female empowerment is inherently multifaceted, even within a single welfare dimension, we frame the selection of survey questions as a multi-task problem, as detailed below.

A.1 Random Forest

Our preferred approach uses the random forest regressor which supports multiple outcome variables and often achieves a high degree of predictive accuracy. It is an ensemble method that fits a large number of decision trees on bootstrap samples (drawn with replacement) from the training set. Essentially, decision trees recursively partition the data using a complex set of if-else decision rules based on cut-off values of features. The cut-off values are chosen to group together the data into subsets (or nodes) such that the variance in the outcome value within each subset is minimized. This ensures that observations with similar output values belong to the same partition. The outcome in the final subsets of the data (known as the leaf nodes) is predicted by taking the average of the outcome in the training data belonging to each node. Note that individual decision trees tend to overfit the training data. Therefore, random forest injects two sources of randomness. First, as discussed above, it builds each tree using a bootstrap sample. In addition, each node in individual decision trees is split based on a random subset of features. The randomness from these two sources results in a low correlation between prediction errors of individual decision trees. We then average these predictions to offset errors and prevent overfitting. We run the random forest regressor 100 times on 50 percent random subsets of the data and use five-fold cross validation within each iteration to choose the best hyperparameters.[17] The remaining 50 percent observations in each iteration are set aside and comprise the out-of-bag sample.

To compute feature importance, we use permutation feature importance introduced by Breiman (2001). For a given feature, this is calculated by measuring the decrease in model score (in this case, R^2) after randomly shuffling that feature. We follow the steps below for every iteration of random forest:

1. Fit the model on 50 percent subsample of the data and, on the remaining 50 percent out-of-bag data denoted as (X^{oob}, Y^{oob}), compute R^2 (or $score^{oob}$) for the random forest regressor:

2. For each feature $x_j \in X$, repeat the following L times (we use L=10):

    - Randomly shuffle (or permute) the feature to create a modified version of the data $(\tilde{X}^{oob}, Y^{oob})$

[17] The two most important hyperparameters for random forest are the number of trees and the number of features in the random subset considered for splitting a node. A larger number of trees improves predictive accuracy at a decreasing rate while increasing computational time. Therefore, we set the number of trees to 1,000 guided by the trade-off between predictive accuracy and computational time. For our second hyperparameter, i.e. the fraction of features to consider when looking for the best split we use the hyperparameter values $[.05, .075, .1, .125, .15]$, and choose the value that gives the best cross-validation score.

- On this modified data and for every repetition $l \in \{1, 2, ..., L\}$ compute R^2 (or $score^{oob}_{j,l}$)

3. The feature importance of variable x_j is given as $F_j = score^{oob} - 1/L \sum_{l=1}^{L} score^{oob}_{j,l}$

We then compute the mean permutation feature importance $\bar{F}_j$ over 100 random forest iterations for each feature. Therefore, permutation feature importance can be interpreted as the average decrease in R^2 on out-of-bag sample when the feature is shuffled. This method overcomes two shortcomings of mean decrease in impurity (MDI)—a commonly used alternative measure for computing feature importance.[18] First, impurity-based feature importance methods inflate the importance of continuous features relative to categorical ones (Strobl *et al.*, 2007). This is because features with high cardinality have more split points, and therefore, are more likely to reduce variance in the child node by chance alone. This is important for us since our predictors include a combination of categorical and continuous variables. Secondly, permutation feature importance can be computed over the out-of-bag sample. This makes it less sensitive to giving high importance to features that are not predictive on the out-of-bag sample when there is overfitting on the training data. Finally, in the interest of maintaining consistency with the alternative methods discussed below and to take into account correlation between features, we rank features based on $\bar{F}_j$ and plot the adjusted R^2 by running a linear regression for each of the outcome measures against the top k predictors.

A.2 Multi-task Lasso Stability Selection

Our second approach uses multi-task Lasso proposed by Argyriou *et al.* (2006), which is an extension of the linear Lasso model but allows us to simultaneously select predictors for multiple outcomes. It jointly selects a sparse set out of J features for multiple (K) regression problems (or tasks) by constraining the set of selected features to be the same across tasks. Consider the linear model:

$$\hat{Y} = X\beta$$

where X represents the set of features (or predictors), Y is the matrix of target variables (or outcomes), and β represents the coefficients. To constrain the set of selected features to be the same across tasks, multi-task Lasso uses a mixed $l_1 l_2$ norm for regularization and solves the following optimization problem:

$$\min_{\beta} ||Y - X\beta||^2_{Fro} + \frac{1}{\lambda}||\beta||_{21}$$

where λ is a constant known as the regularization parameter and Fro indicates the Frobenius norm:

$$||Y - X\beta||_{Fro} = \sqrt{\sum_{k=1}^{K}\sum_{i=1}^{N}(y_{i,k} - x_i\beta_k)^2}$$

where i indexes an observation, k indexes the task, and j indexes the parameter. In our case, the outcomes are different measures of relative welfare within a dimension. The $l_1 l_2$ mixed norm is given

[18] MDI computes feature importance by measuring the sum of decrease in within-node variance of outcome (or impurity) over all nodes split using that feature, averaged over all trees in the forest. In other words, it is the difference in the impurity of a node and the weighted (by fraction of samples reaching that node) sum of impurity of child nodes obtained by splits on a feature.

by:

$$||\beta||_{21} = \sum_{j=1}^{J} \sqrt{\sum_{k=1}^{K} \beta_{j,k}^2}$$

We combine this with the stability selection algorithm wherein we repeatedly run multi-task Lasso on sub-samples of our data. Stability Selection was proposed by Meinshausen & Bühlmann (2010) and has previously been applied by Jayachandran *et al.* (2023) using Lasso. We instead use multi-task Lasso and run the algorithm separately for the different welfare dimensions using the following steps. First, we take a 50% sample of observations without replacement. We then run multi-task Lasso with five-fold cross-validation to choose the regularization parameter.[19] This gives us a sparse set of selected predictors, i.e. for which the coefficients do not shrink to 0. We then repeat this process for 1,000 iterations and order the predictors based on the number of times they are selected for each dimension. Finally, we regress each of the outcome measures on the top k predictors and record the adjusted R^2 obtained. We also check the sensitivity of our results using Lasso stability selection, which does not impose the restriction of selecting the same set of features across different outcomes within a dimension.

A.3 Group Lasso Stability Selection

As described in Section 3.2, we re-code responses to questions with categorical answers into multiple binary variables. Therefore, to identify which questions (as opposed to which variables) are predictive of women's empowerment, we also use group lasso (Yuan & Lin, 2006). To implement this, we group together indicator variables created from a single (or merged) categorical question while keeping questions with continuous/ranked responses and binary responses as part of distinct groups. In other words, each unique survey question is mapped to a group of predictors resulting in G groups. We solve a similar optimization problem as before:

$$\min_{\beta} ||Y - \sum_{g=1}^{G}[X_g \beta_g]||_{Fro}^2 + \frac{1}{\lambda}||\beta||_{21}$$

For notational convenience, we separately include coefficients for group g in the Frobenius norm:

$$||Y - \sum_{g=1}^{G}[X_g \beta_g]||_{Fro} = \sqrt{\sum_{k=1}^{K}\sum_{i=1}^{N}(y_{i,k} - \sum_{g=1}^{G}[x_{i,g}\beta_{g,k}])^2}$$

However, the penalty or regularization term is slightly modified to impose group-wise sparsity:

$$||\beta||_{21} = \sum_{g=1}^{G} \sqrt{\rho_g} \sqrt{\sum_{k=1}^{K}\sum_{j=1}^{\rho_g} \beta_{j,g,k}^2}$$

[19] For pre-processing, we remove duplicate (or perfectly correlated) predictors and features with very little variation (i.e. those that vary less than 5 times in our data). We also remove outliers—observations for which relative calorie consumption is 0 or 1. To ensure that the continuous variables do not dominate the objective function, we scale them to a range between 0 and 1 before using the coordinate descent algorithm to fit the model.

where ρ_g is the number of predictors in group g such that $\sum_{g=1}^{G} \rho_g = J$. This penalty term jointly selects a sparse set of groups and ensures that all the predictors created out of a single categorical variable are included and excluded jointly, while restricting the set of groups to be the same across tasks. Note that if each covariate belongs to its own group, then this would reduce to multi-task lasso. We follow the same pre-processing steps and combine group lasso with the stability selection procedure described in the previous sub-section, and rank variables based on their selection frequency.

B Classifying Questions into Power Domains using Sentence Transformer

In the main analysis, we rely on a manual categorization of survey questions into *power domains*. To assess the validity of this approach, we also cluster the questions into domains based on text embeddings obtained using a sentence transformer model and compare the classification with the manual categories. The procedure involves the following four steps:

1. **Question-domain pairs.** We first use the BIHS questionnaire to construct a list of questions (e.g., "Did you participate in [ACTIVITY] in the past 12 months? How much input did you have in making decisions about [ACTIVITY]?") along with the domain to which each variable refers (e.g., "Fishing or fish culture"). We then remove duplicates to retain 147 unique question-domain combinations, so that questions with more response categories do not receive a disproportionate weight during clustering.

2. **Embedding generation.** We generate embeddings (vector representations) for each question (v_q) and its associated domain (v_g) using a `sentence transformer` model (Reimers & Gurevych, 2019). Conventional transformers such as BERT or RoBERTa are less suitable for this task, as they obtain sentence representations by averaging contextualized word embeddings. This limits their effectiveness for applications that depend on semantic similarity. The sentence-transformer framework addresses this by fine-tuning transformer models to produce sentence-level embeddings that capture the overall semantic meaning. Specifically, we use the pre-trained `all-mpnet-base-v2` model, a fine-tuned variant of Microsoft's *mpnet-base* (Song *et al.*, 2020), trained on over a billion sentence pairs from diverse sources (e.g., academic papers, Wikipedia, Reddit, and Stack Exchange). This model encodes text into a 768-dimensional vector space and ranks among the top-performing models for *semantic textual similarity*.

   Given a set of N sentence pairs (v_i^1, v_i^2), the model computes embeddings $\vec{v}_i^1$ and $\vec{v}_i^2$ for $i \in \{1, ..., N\}$ by taking the mean of contextualized word embeddings using a conventional transformer. It then calculates the cosine similarity $cos(\vec{v}_i^1, \vec{v}_j^2)$ for all possible sentence pairs, forming a similarity matrix M whose elements $M_{i,j}$ represent the pairwise cosine similarity between sentences v_i^1 and v_j^2. Finally, the embeddings are fine-tuned by minimizing a symmetric cross-entropy loss that maximizes the similarity between true sentence pairs relative to all others:

$$l = -\frac{1}{2N}\sum_{i=1}^{M}\left(log\frac{exp(cos(\vec{v}_i^1, \vec{v}_i^2))}{\sum_{j=1}^{N} exp(cos(\vec{v}_i^1, \vec{v}_j^2))} + log\frac{cos(\vec{v}_i^2, \vec{v}_i^1)}{\sum_{j=1}^{N} exp(cos(\vec{v}_i^2, \vec{v}_j^1))}\right)$$

In our context, this ensures that conceptually related questions—such as those reflecting similar aspects of bargaining power—are located closer together in the embedding space and form smaller angles between their corresponding vectors, while dissimilar questions are positioned further apart. Semantic similarity can then be quantified using cosine similarity. Furthermore, sentence transformers are highly computationally efficient, operating orders of magnitude faster than conventional transformer models.

3. **Composite embedding.** For each question, we construct a composite embedding

$$v = (1-w)v_q + wv_g,$$

where w denotes the relative weight placed on the domain. In the baseline specification, we set $w = 0.6$, reflecting the idea that domains should play a more central role in classification while still retaining information from the questions. If the domain is missing, we set $w = 0$. This choice yields results most consistent with our manual categorization (as discussed below). For comparison, we also consider $w = 0.9$, which appears to perform slightly better on visual inspection. Finally, we systematically vary w from 0 to 1 in increments of 0.05 to examine the sensitivity of the results to the relative importance placed on domains versus questions.

4. **Clustering.** We first reduce the 768-dimensional embeddings to 15 dimensions using Uniform Manifold Approximation and Projection (UMAP)—a non-linear dimensionality reduction method that preserves the underlying topological structure of the data (McInnes *et al.*, 2018).[20] This dimension reduction step improves the effectiveness of clustering algorithms, which tend to perform poorly in very high-dimensional spaces.[21] We then apply the k-means clustering algorithm to partition the survey questions into 10 domains with the objective of minimizing the within-cluster sum of squared Euclidean distances (inertia). The algorithm begins by randomly initializing cluster centroids and then proceeds iteratively: each point is assigned to the nearest centroid, and the centroids are updated as the mean of the points in each cluster. This process continues until assignments stabilize.[22] We obtain similar results using HDBSCAN, a hierarchical, density-based clustering algorithm (Campello *et al.*, 2013).[23]

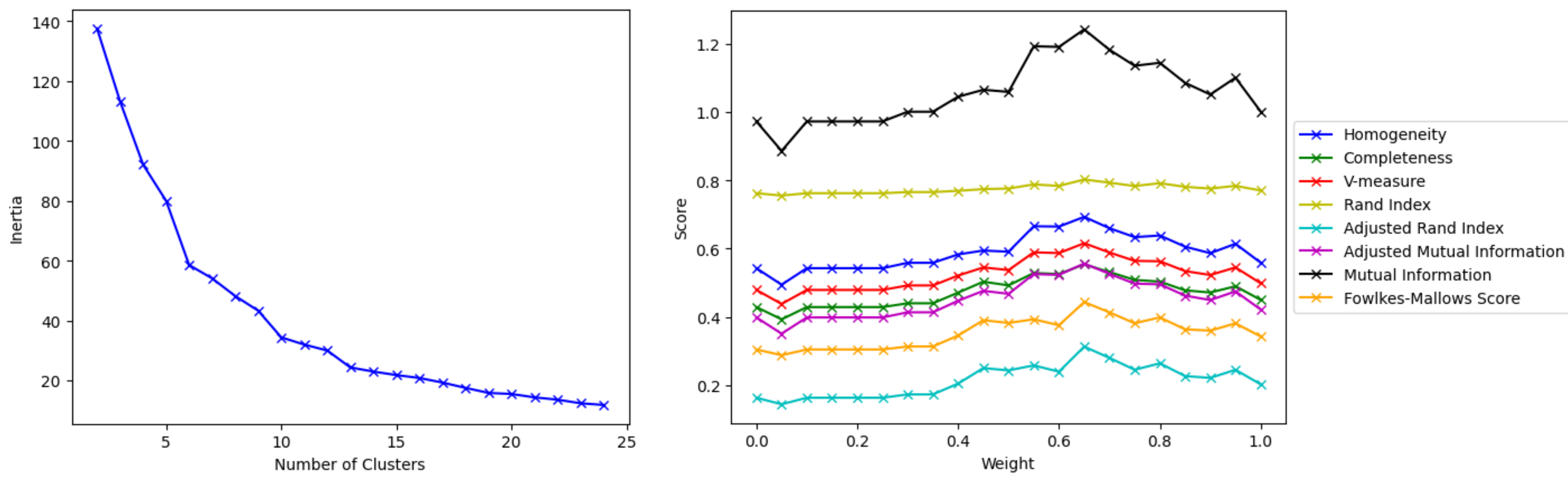
[20]The key hyperparameter in UMAP is the number of nearest neighbors. Since our dataset contains 147 questions, each question can have at most 146 neighbors. We therefore set this parameter to 146 to capture the global structure of the data. We also set the minimum distance between points to 0, which produces a denser low-dimensional representation more suitable for clustering, and use cosine distance as the similarity metric.

[21]This "curse of dimensionality" arises because Euclidean distances lose their discriminatory power as dimensionality increases: the gap between the nearest and farthest points shrinks, so observations appear almost equally distant. At the same time, the data become sparse, leaving large empty regions that make it difficult for algorithms to detect dense, well-defined clusters.

[22]To determine the number of clusters, we use the elbow method by plotting inertia as the number of clusters increases from 2 to 25 and selecting the point where the curve flattens, indicating diminishing returns from additional clusters. We depict this in Figure A1, Panel (a).

[23]For HDBSCAN, we set the *minimum cluster size* to 4 (ensuring that each domain contains at least four questions), *minimum samples* to 1 (so that even a single point in a dense region is treated as a core point and fewer questions are discarded as "noise"), and $\epsilon = 0.65$, so that clusters closer than this threshold in Euclidean distance within the UMAP space are not split further. For an accessible introduction to these methods and their application in economics, see Chaturvedi *et al.* (2024), who use them to analyze online job ad texts.

Figure A1: Domain Classification



(a) Inertia vs. # of clusters ($w = 0.6$)

(b) Cluster evaluation metrics

Note: Panel (a) shows the elbow method, where the optimal number of clusters is chosen at the point where inertia flattens. Panel (b) compares the clusters against the manually constructed domains, using standard metrics including homogeneity, completeness, V-measure, Rand index, adjusted Rand index, adjusted and unadjusted mutual information, and Fowlkes–Mallows score, for 10 clusters across different values of w.

We compare the clusters against the manually constructed domains using standard measures, including homogeneity, completeness, V-measure, Rand index, adjusted Rand index, mutual information (adjusted and unadjusted), and the Fowlkes–Mallows score. The results are shown in Figure A1, Panel (b) for 10 clusters for different values of w. Across specifications, the clustering results align closely with the manual classification, suggesting that the findings reported in the main analysis are not sensitive to the particular classification method employed.[24] We find that the concordance with the manual categorization tends to be the highest for weight w around 0.6–0.7. For example, with $w = 0.6$, the homogeneity and completeness scores are 0.66 and 0.53, respectively. In practice, this implies that about 66% of the questions within a cluster belong to the same manually classified domain, while 53% of the questions in a given manual domain are grouped into the same cluster.

[24] For detailed definitions of these metrics, see `https://scikit-learn.org/stable/modules/clustering.html`.

C Data Construction

Below we describe the outcome variables for our prediction exercises. Distributions of the respective outcomes are presented in Figure A2, with corresponding descriptive statistics in Table A2.

Health Outcomes: We use four measures of health as outcome variables in our analyses, each intended to capture a distinct dimension of individual health and well-being: an illness index, a disability index, a domestic violence index, and body mass index (BMI). The illness index is constructed using principal component analysis (PCA) and is designed to summarize short-term health problems. Specifically, the PCA incorporates separate indicators for whether the respondent was sick or lost weight during the preceding four weeks, as well as the number of days the respondent was sick over the same period. The first principal component is driven primarily by illness in the previous four weeks, days sick, and recent weight loss, with component coefficients of 0.652, 0.611, and 0.449, respectively. In contrast, underweight status has a coefficient close to zero on the first component (0.015) and instead loads strongly on the second component (0.967).

The disability index is also constructed using a PCA and summarizes long-term functional limitations across several domains, including the respondent's ability to stand, walk, carry heavy objects, speak, see, and move. The first principal component is driven most strongly by difficulty walking five kilometers, carrying objects, and standing, with component coefficients of 0.509, 0.503, and 0.488, respectively. Difficulty seeing also contributes substantially to the index, with a coefficient of 0.384, while difficulty hearing has a smaller positive coefficient of 0.199. Difficulties related to paralysis, bodily functioning, and speaking have comparatively lower coefficients of 0.154, 0.149, and 0.125. Because all first-component coefficients are positive, higher values of the resulting principal-component score indicate greater overall disability, with the index primarily capturing limitations in mobility and physical functioning.

Since the outcomes summarized using a principal component analysis can be negative, we rescale the resulting score to lie between zero and one using an inverse-logit transformation to facilitate the ratio calculation. Couple-relative outcomes are then constructed as the wife's value divided by the sum of the wife's and husband's values. Higher values of the resulting ratio indicate the wife is in relatively worse health than her husband.

The domestic violence index is constructed from Module Z4 of the BIHS questionnaire, "Domestic Violence, Abuse and Threats". This module records whether the respondent experienced different forms of violence, abuse, or threats during the previous year. These include threats by the husband to divorce the respondent or take another wife, verbal abuse by the husband or by other adult household members, and physical abuse by the husband or by other adult male or female household members. For each form of abuse, the questionnaire distinguishes between events that occurred often, occurred sometimes, did not occur, were not applicable, or for which the respondent declined to answer. For cases of physical abuse, the module further records the severity of harm through information on the type of injury sustained, including cuts, bruises, aches, eye injuries, sprains, dislocations, burns, deep wounds, broken bones, broken teeth, and other serious injuries. The module also captures more extreme forms of coercion and exclusion, including whether the respondent was ever threatened with being forced to leave the household, whether she was ever forcefully sent out of the household, who brought her back if this occurred, and why she returned if she came back herself. We aggregate these items using a principal component analysis to construct an index that reflects both the incidence and severity of domestic violence, abuse, and threats. Higher values indicate more domestic violence.

Time Use Outcomes: Our primary measures of time use are hours spent in market work and domestic work. We construct measures of market work and domestic work using a detailed 24-hour time-use diary for male and female respondents. The module asks enumerators to record the respondent's activities over the last complete 24 hours, beginning at 4:00 a.m. on the previous day and ending at 3:59 a.m. on the day of the interview. Time is recorded in 15-minute intervals. The questionnaire records a broad set of activities, including sleeping, eating, personal care, schooling, paid employment, own-business work, farming or fishing, shopping or obtaining services, weaving or textile care, cooking, domestic work, care for children, adults, or the elderly, commuting, travelling, leisure, social activities, religious activities, and other activities.

We use these time-use data to construct two measures of labor allocation. Market work is defined as time spent in income-generating or market-oriented activities, including work as an employee, own-business work, farming or fishing, and commuting or travelling associated with these activities. Domestic work is defined as time spent in unpaid household production and care responsibilities, including cooking, domestic work, shopping or obtaining services, weaving, sewing or textile care, and care for children, adults, or the elderly. For each respondent, we aggregate the relevant 15-minute intervals within each category to obtain total time spent in market work and total time spent in domestic work over the previous day. These measures provide individual-level summaries of labor allocation across paid or productive work and unpaid household responsibilities. With market and domestic work for each spouse, we then construct the final outcome measure as the ratio of the wife's hours worked divided by the sum of the wife and husband's hours. If neither spouse works, we code this as 0.5.

We construct our time-use satisfaction index using responses from Module WE6B of the BIHS questionnaire, "Satisfaction with Time Allocation." The module's purpose is to measure respondents' time use and their subjective satisfaction with different dimensions of that time allocation. In particular, we use questions that ask respondents to rate their satisfaction on a scale from 1 to 10, where 1 indicates that the respondent is not satisfied, 10 indicates that the respondent is very satisfied, and 5 represents a neutral response. The question elicits satisfaction across six domains: the distribution of work duties within the household; available time for leisure activities, such as watching television, listening to the radio, seeing movies, or doing sports; contact with friends or relatives; the possibility of going to places outside the village; the respondent's power to make important decisions that change the course of life; and overall life satisfaction. We aggregate these six items using principal component analysis (PCA) to construct a summary measure of subjective satisfaction. The resulting index captures variation in respondents' perceived well-being across dimensions related to time use.

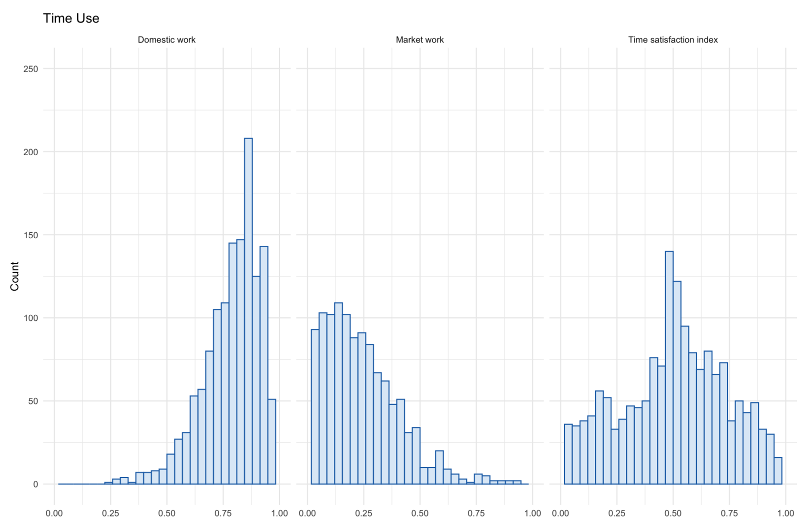
Food Outcomes: We use several measures of food consumption that are designed to capture both the quantity and quality of what they eat each day. The food-consumption data come from the individual dietary module of the Bangladesh Integrated Household Survey. A female enumerator visited each household and interviewed the woman primarily responsible for preparing food. Using a 24-hour recall, the respondent listed the meals and food items consumed by the household on the preceding day. For prepared dishes, the survey recorded the raw and cooked weights of each ingredient. For example, a curry was separated into its constituent quantities of fish, potatoes, onions, and other ingredients. The respondent then reported the share of each meal consumed by every household member. The survey also accounted for food given to guests or animals, leftovers, meals eaten outside the home, and household members who did not participate in a meal.

Individual food quantities can be constructed by allocating each dish and its ingredients to household members according to their reported shares of the meal. The monetary value of an individual's food consumption can then be calculated by applying the corresponding food prices or household-reported values to the quantities assigned to that person and summing across all foods consumed during the recall day. To make the one-day observation more representative of usual consumption, households were asked about the most recent typical day whenever the preceding day was a special occasion. Interviews were not conducted during Ramadan, and the paper excludes households that fed guests on the recall day.

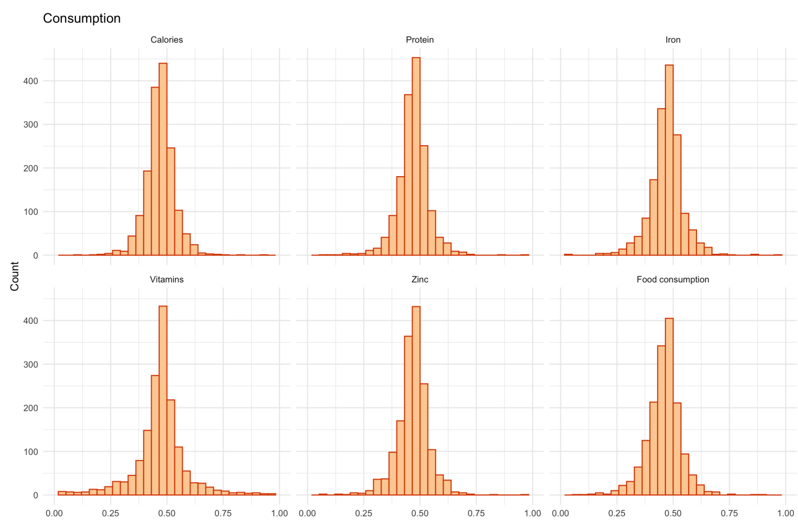
Calorie and nutrient intake can be obtained by matching the individual quantities of each food or ingredient to food-composition information and summing its caloric and nutrient content across all items. For our outcomes, we focus on daily calorie, protein, iron, vitamin, and zinc consumption as indicators of the nutritional quality. For our outcome measures, we take the ratio of women's consumption within these dimensions, divided by the sum of men's and women's consumption.

D Additional Figures

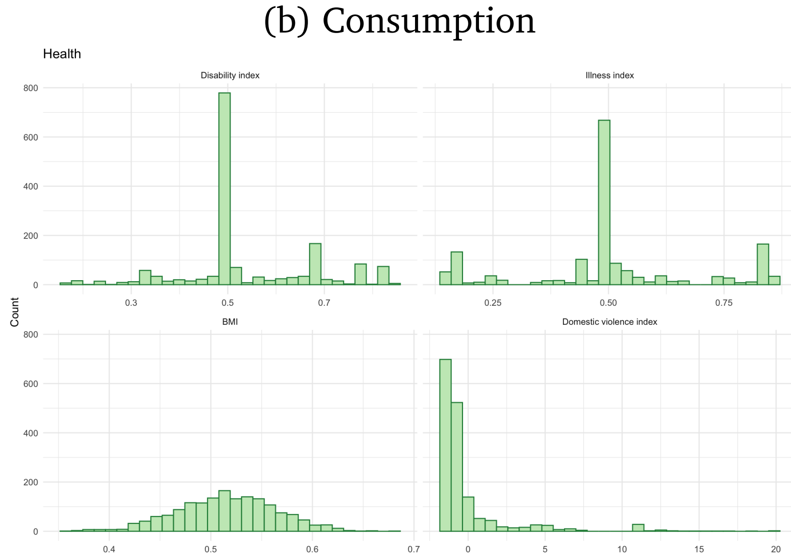
Figure A2: Intra-household Relative Outcomes in Various Domains



(a) Time use



(b) Consumption



(c) Health

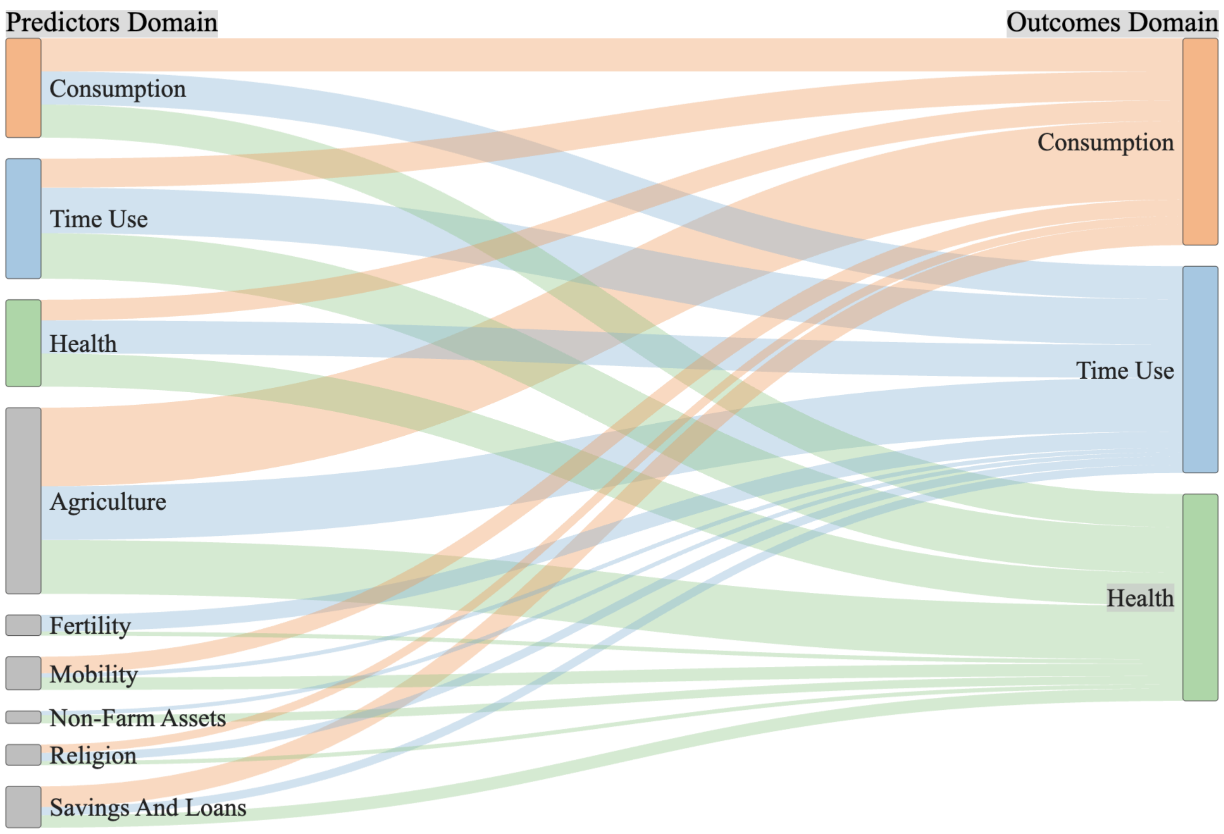
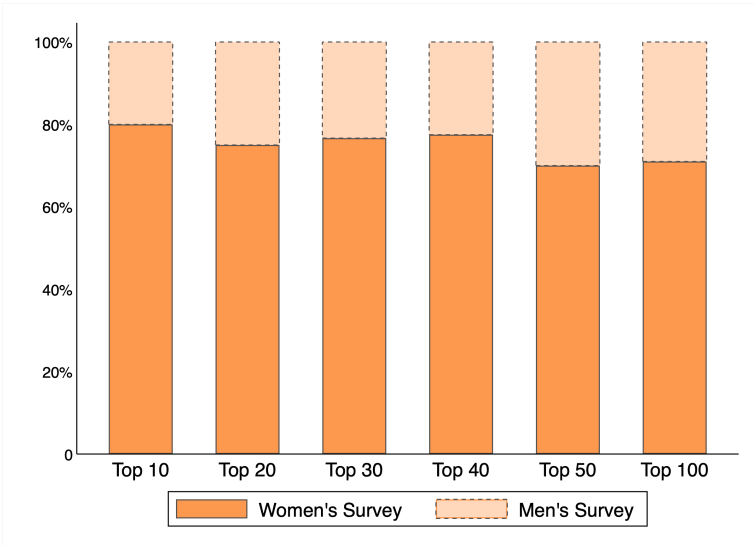


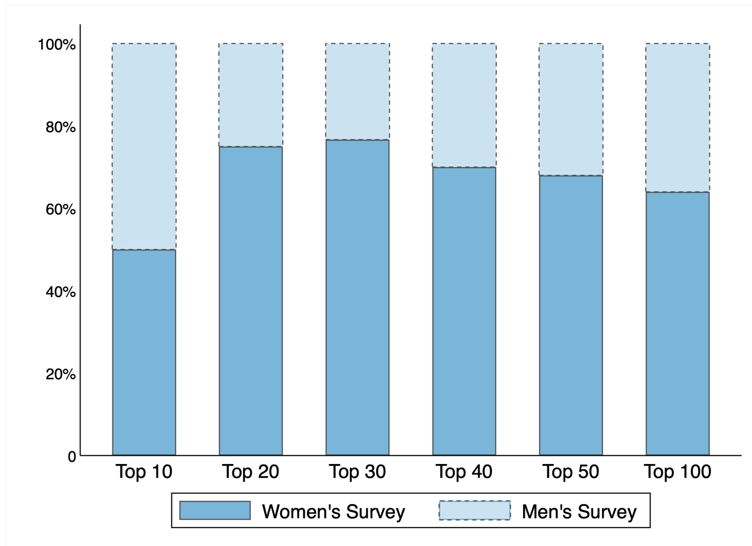
Figure A3: Multiple Dimensions of Intra-household Power (Details)

NOTES: The Sankey diagram illustrates the distribution of the top 50 survey questions (ranked by random forest permutation feature importance) across power domains and outcome dimensions. The width of each flow represents the share of variables from a given power domain that are among the top 50 predictors of women's relative well-being in consumption, time use, and health.

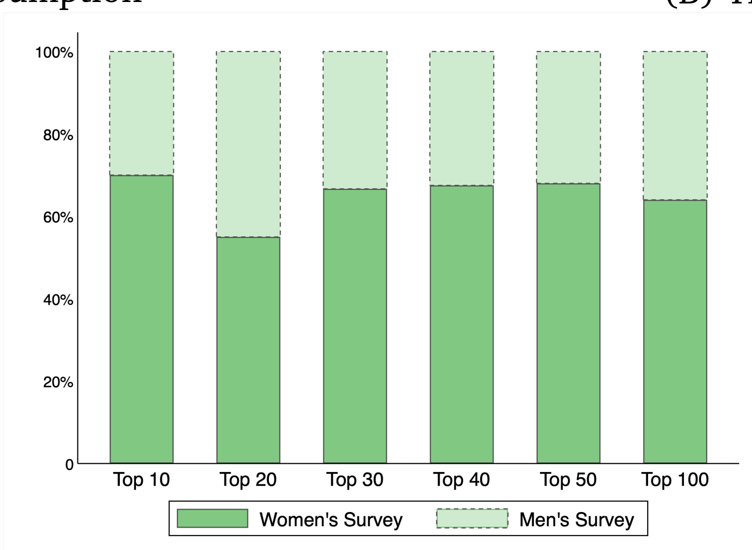
Figure A4: Share of Top Predictors by Respondent Gender



(A) Consumption



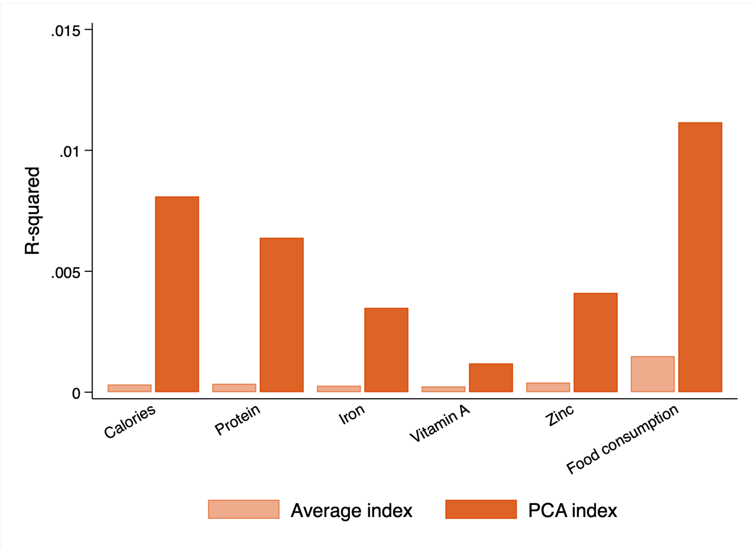
(B) Time Use



(C) Health

NOTES: Random Forest Estimation. The dark bars represent the share of the top X questions ranked by feature importance that are asked to women, with the light bars representing the share asked of men.

Figure A5: Predictive Power of Single-Index Measures by Outcome



(A) Consumption

(B) Time Use

(C) Health

NOTES: The figure reports outcome-specific R^2 values from regressions of each relative wellbeing outcome on pooled single-index measures constructed from the full set of power-related survey indicators used as predictors. The average index is the equal-weighted average of standardized indicators. The PCA index is the first principal component of the standardized indicators. No covariates are included.

E Additional Tables

Table A1: Module Overview

Module	Description	Questions for Women	Questions for Men
Women's Status			
Z1	Work Earnings and Expense	11	-
Z2	Freedom of Mobility	10	-
Women's Empowerment in Agriculture Index (WEAI)			
WEA	Ability to Be Interviewed Alone	1	1
WE2	Role in Household Decision-Making Around Production and Income Generation	16	16
WE3A	Access to Productive Capital	54	48
WE3D	Access to Loans	21	25
WE5A	Decision Making in the Household	21	15
WE5B	Decision Making Vignettes	12	12
WE5C	Motivation for Decision Making	32	32
WE6B	Satisfaction with Time Allocation	2	2
Total		172	143

Note: Only survey questions used in the estimation are included in the table.

Table A2: Target Variables

	Descriptive Statistics				Explanatory Power of Demographics
	Mean	Median	Std. Dev	Obs.	R^2
Time Use					
Domestic Work	0.822	0.844	0.143	1620	0.023
Market Work	0.208	0.151	0.232	1620	0.075
Time Use Satisfaction	0.514	0.508	0.226	1620	0.055
Food Consumption					
Food Consumption	0.461	0.465	0.080	1620	0.024
Caloric Intake	0.471	0.471	0.069	1620	0.019
Protein Intake	0.471	0.472	0.073	1620	0.018
Iron Intake	0.472	0.474	0.078	1620	0.019
Vitamin Intake	0.472	0.477	0.118	1620	0.019
Zinc Intake	0.470	0.472	0.076	1620	0.017
Health					
Health Index	0.540	0.500	0.133	1620	0.020
Illness Index	0.505	0.500	0.185	1620	0.017
BMI	0.516	0.516	0.049	1620	0.037
Domestic Violence Index	0.000	-0.612	2.599	1620	0.014

Note: BIHS data. Variables are ratios computed as $\frac{y_w}{y_w+y_h}$ where y_w and y_h denote the values of some variable for the wife and husband, respectively, with the exception of the domestic violence index which is only computed for women. Time Use Satisfaction is computed using a principal components analysis of several survey questions on stated satisfaction. The R^2 refers to a regression of the outcome variable on standard demographic information of the both spouses including their education, age, age^2, literacy, and the number of children in the household.

Table A3: Top Predictors of Spousal Relative Consumption

Question	Answer	Feature Importance
Your actions with respect to how to express religious faith are motivated by a desire to avoid blame or so that other people speak well of you? Can you tell me whether it is entirely true, somewhat true, not very true or never true?^{W}	Somewhat true	0.0006
Did you participate in raising poultry in the past 12 months, and if so, how much input did you have on income generated from it?^{M}	Yes, I had input into very few decisions	0.0003
Does anyone in your household currently have this chickens, ducks, turkeys or pigeons, and if so, who contributes most regarding a decision for a new purchase of it?^{W}	Yes, spouse	0.0003
"[PERSON'S NAME] can't raise any livestock other than what she has. These are all that's available." Are you like this person?^{W}	Yes/no	0.0003
Regarding the amount of sleep you got last night, was that:^{W}	Less than average	0.0003
Has anyone in your household taken any loans or borrowed cash/in-kind from an informal lender in the past 12 months, and if so, who makes the decision on what to do with the money/item?^{W}	Yes, self	0.0003
When decisions are made regarding what types of crops to grow, who is it that normally takes the decision, and if not you, to what extent do you think you can make decisions if you wanted to?^{W}	Spouse, to a medium extent	0.0003
Has anyone in your household taken any loans or borrowed cash/in-kind from an informal lender in the past 12 months, and if so, who made the decision to borrow from them?^{M}	Yes, spouse	0.0003
Your actions with respect to how to express your religious faith are motivated by a desire to avoid punishment or gain reward? Can you tell me whether it is entirely true, somewhat true, not very true or never true?^{W}	Somewhat true	0.0003
Do you yourself control the money needed to buy food from the market?^{W}	Yes/No	0.0003

Note: Random forest. Most important predictor variables ranked by feature importance. We report the survey question and categorical response that forms the predictor. Questions are edited for conciseness and clarity. Superscripts in the question column indicate whether the respondent is a man (M) or woman (W).

Table A4: Top Predictors of Spousal Relative Health

Question	Answer	Feature Importance
Does anyone in your household currently have large livestock, and if so, who contributes most regarding a decision for a new purchase of it?^{M}	Yes, spouse	0.0022
Your actions with respect to whether and how to express your religious faith are motivated by and reflect your own values and/or interests? Can you tell me whether it is entirely true, somewhat true, not very true or never true?^{W}	Somewhat true	0.0021
Your actions with respect to whether and how to express your religious faith are motivated by and reflect your own values and/or interests? Can you tell me whether it is entirely true, somewhat true, not very true or never true?^{W}	Always true	0.0016
Your actions with respect to how to protect your self from violence are motivated by a desire to avoid punishment or gain reward? Can you tell me whether it is entirely true, somewhat true, not very true or never true?^{M}	Somewhat true	0.0014
Did you participate in wage or salary employment in the last 12 months, and if so, how much input did you have in decisions?^{W}	Yes, input into most decisions	0.0014
Your actions with respect to what to do if you have a serious health problem are motivated by a desire to avoid punishment or gain reward? Can you tell me whether it is entirely true, somewhat true, not very true or never true?^{M}	Somewhat true	0.0011
Your actions with respect to how to protect yourself from violence are motivated by a desire to avoid blame or so that other people speak well of you? Can you tell me whether it is entirely true, somewhat true, not very true or never true?^{M}	Somewhat true	0.0011
Your actions with respect to how to protect yourself from violence are motivated by and reflect your own values and/or interests? Can you tell me whether it is entirely true, somewhat true, not very true or never true?^{W}	Always true	0.0010
When decisions are made regarding major household expenditures of household life, who is it that normally takes the decision, and if not you, to what extent do you think you can make decisions if you wanted to?^{M}	NA	0.0010
Did you participate in wage or salary employment in the last 12 months, and if so, how much input did you have on income generated from it?^{W}	Yes, input into most decisions	0.0010

Note: Random forest. Most important predictor variables ranked by feature importance. We report the survey question and categorical response that forms the predictor. Questions are edited for conciseness and clarity. Superscripts in the question column indicate whether the respondent is a man (M) or woman (W).

Table A5: Top Predictors of Spousal Relative Time Use

Question	Answer	Feature Importance
Your actions with respect to minor household expenditures are motivated by a desire to avoid punishment or gain reward? Can you tell me whether it is entirely true, somewhat true, not very true or never true?^{W}	Always true	0.0022
Your actions with respect to what to do if you have a serious illness are motivated by a desire to avoid blame or so that other people speak well of you? Can you tell me whether it is entirely true, somewhat true, not very true or never true?^{W}	Always true	0.0019
Your actions with respect to how to protect yourself from violence are motivated by a desire to avoid blame or so that other people speak well of you? Can you tell me whether it is entirely true, somewhat true, not very true or never true?^{W}	Always true	0.0017
Does anyone in your household have a house? If so, who decides whether to sell it?^{M}	No, NA	0.0015
Who decides how to spend money on housing?^{W}	Yourself	0.0013
When decisions are made regarding minor household expenditures, who is it that normally takes the decision, and if not you, to what extent do you think you can make decisions if you wanted to?^{M}	Self, NA	0.0011
Does anyone in your household have large consumer durables? If so, who decides whether to give it away?^{M}	No, NA	0.0009
When decisions are made regarding major household expenditures, who is it that normally takes the decision, and if not you, to what extent do you think you can make decisions if you wanted to?^{W}	Husband, not at all	0.0007
Who decides how to spend money on healthcare?^{W}	Yourself	0.0007
Your actions with respect to whether or not to use family planning are motivated by a desire to avoid blame or so that other people speak well of you? Can you tell me whether it is entirely true, somewhat true, not very true or never true?^{W}	Always true	0.0007

Note: Random forest. Most important predictor variables ranked by feature importance. We report the survey question and categorical response that forms the predictor. Questions are edited for conciseness and clarity. Superscripts in the question column indicate whether the respondent is a man (M) or woman (W).